MILITARY SPECIFICATION

TEST set, INVERTER, CONVERTER AND MOTOR CHASSIS ASSEMBLY: 11748238

This specification is approved for use by l ll Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of Inverter, Converter and Motor Chassis Assembly Test Set, hereinafter referred to as the “Chassis Assembly” which forms a part of the Electronics System Test Set type AN/TSM-100.

2. APPLICABLE DOCUMENTS

2,1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this Specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-F-13926	Fire Control Material: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Material: Packaging of
MIL-T-43435	Tape: Impregnated, Lacing and Tying
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction; M18 Manufacturing Processes and Inspection of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-810	Enviromental Test Methods
MIL-STD-454	General Requirements for Electronic Equipment

FSC 4931

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.

DRAWINGS

U.S. Army, Frankford Arsenal

F11748238	Chassis Assembly, inverter Converter l nd Motor
F10554035	Schematic Diagram, Electrical, Chassis Assembly, Inverter, Converter l nd Motor
B10554604	Wire Runs List, Chassis Assy. Inv. - Conv.

PACKAGING DATA SHEET

11748238	Packaging of Test Set, Chassis Assembly

(Copies of specification, standards, drawings and packaging data sheets, required by suppliers in connection wiht specific procurement functions should be obtained from the procuring l ctivity or l s directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The test set shall be manufacture in l ccordance with Drawing F11748238 and drawings pertaining thereto.

3.2 General specifications.- Where applicable, the chassis assembly shall meet the requirements of MIL-F-13926 and MIL-C-46357 as follows:

3.2.1 MIL-F-13926.

a. Order of precedence.
b. Dimensions and tolerances.
c. Inorganic protective surface finishings.
d. Part identification and marking,
e. Workmanship.
f. Soldering electrical connections.

3.2.2 MIL-C-46357.

a. Etched circuit assemblies
b. wiring
c, Threaded fasteners
d. Marking
e. Finishes

3.3 Environmental service condition.

3.3.1 Vibration (nonoperating).- The test set shall be subject to a simple harmonic motion having excursion as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.

Frequency (Hz ± 2 percent)	Excursion (double amplitude) (inches ± 10 percent)
5 to 15	0.06
15 to 25	0,04
25 to 55	0.02

The frequency shall be varied logarithmically between 5 and 55 cycles per second (cps) within 15 minutes and returned to 5 cps in 15 minutes. The motion shall be applied to the test set for a period of 30 minutes in each direction of the 3 mutually perpendicular axes (total of 90 minutes), relative to its normal operational position. Subsequent to vibration, the test set shall not be damaged or adversely affected and shall meet the electrical requirements of 3.4.

3.3.2 Temperatures.

3.3.2.1 Operating - The chassis assembly shall meet the requirements of 3.4 while exposed to temperatures of +32% and +131°F.

3.3.2.2 Storage.- The chassis assembly sha 1 meet the requirements of 3.4 at a temperature of between 60°F and 90°F after having been exposed to each temperature of -80°F and +185°F.

3.3.3 Humidity.- The chassis assembly shall function as specified in 3.4 and shall not suffer any damage or deterioration which shall impair satisfactory performance after being subjected to ten consecutive humidity cycles at relative humidities between 85 and 95 percent and temperatures from 82°F to 160% over a period not less than 240 hours.

3.3,4 Shock.- The chaasis assembly shall show no evidence of damage or be l dversely affected after being subjected to l total of 18 shocks consisting of three mutually perpendicular axes. Each shock shall have an acceleration of 15 gravity units (G's) and a pulse duration of 11±1 millisecond.

3.3.5 Altitude.

3.3.5.1 Operating.- The chassis assembly shall not be damaged or l dversely affected when exposed for l period of not less than four hours to conditions encountered at an altitude of 10,000 feet l bove sea level. During and subsequent to such exposure the test set shall meet the requirement of 3.4.

3.3.5.2 Nonoperating.- The chassis assembly shall meet the requirements of 3.4 after exposure to conditions encountered at 50,000 feet above sea level for a period of not less than one hour. The chassis l soambly shall not be damaged or be adversely affected due to such exposure.

3.3.6 Sand and dust.- The equipment shall withstand, in both l n operating and nonoperating condition, exposure to sand and dust particles as encountered in desert areas.

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:33Z
Check the source to verify that this is the current version before use.

3.3.7 Salt spray.- Equipment shall withstand in both an operating and nonoperating condition, exposure to salt-sea atmosphere.

3.3.8 Fungus-inert materials.- The requirements for fungus-inert materials shall be in accordance with MIL-STD-454, Requirement 4,

3.4 Electrical.

3.4.1 Continuity.- The circuitry shall be in accordance with Wire Runs List B10554604 and electrical schematic drawing F10554035. The wire runs list shall take precedence over the schematic drawing. The symbols referenced in subsequent paragraphs pertain to this schematic and are used therein.

3.4.2 Functional.

3.4.2.1 Motor Assembly,

3.4.2.1.1 Conditions of operation,- Unless otherwise specified, the requirements of 3.4.2.1 shall be met wiht the test configuration shown in Figure 1,

3.4.2.1.2 Counterclockwise output.- When S-2 is in the CCU position, and the Servo Load Select Switch (S-1) set to the position specified in Table I, the d.c. voltages stated in Table I shall be present l t the Servo Load (Tp-2, Tp-3) , Tach (Tp-6, Tp-7), anti Tp-4 and Tp-5 test points respectively.

TABLE I

SERVO AMPLIFIER TEST POINT VOLTAGES

LOAD SELECT SWITCH POSITION	TEST POINT VOLTAGES SERVO LOAD Tp-2, Tp-3	TACH (a) Tp-6, Tp-7	Tp-4, Tp-5
1/3	11.0±2.0mV	75V min. (b)	25±2V
2/3	25.0±5.0mV	45 ± 5V	23 ± 2V
Full	34.0±7.0mV	35 ± 5V	22 ± 2V

(a) NOTE: Tach voltages are negative for CCW rotation and positive for CW rotation.

(b) NOTE: If voltage less than 75V, adjust R8 until tach voltage greater than 75V.

3.4.2.1.3 Clockwise output.- With the conditions of 3.4.2.1.2 duplicated except S-2 shall be in the CW position, the test point voltages shall be as stated in Table I for the specified Load Select Stitch position (S-l).

3.4.2.2 Resistor loads.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.

3.4.2.2.1 Conditions of operations.- Unless otherwise specified the requirements of 3.4.2.2 shall be met with the conditions as specified in Table II.

3.4.2.2.2 Resistor load outputs.- When the resistor loads are supplied with the inputs as specified in Table II, the resistor loads shall yield the outputs specified in Table II.

TABLE II

J1 Terminals	D.C. Input Power (Voltage in series with resistor)	D.C. Output Voltage
26 & 32	41.0V ± 1%; 7 ohms, ± 1%, 30W	20.5V ± 2%
21 & 32	227.9V±1%; 649 ohms, ± 1%, 20W	113.9V ± 2%
6 & 27	289.4V±1%; 698 ohms, ± 1%, 30W	144.7V ± 2%
16 & 32	37.6V±1%; 11.8 ohms, ±1%, 30W	18.8V ± 2%
19 & 32	37.6V±1%; 11.8 ohms, ± 1%, 30W	18.8V ± 2%
1 & 32	219.6V±1%; 201 ohms, ±1%, 30W	109.8V ± 2%
22 & 32	219.6V±1%; 201 ohms, ±1%, 30W	109.8V ± 2%
12 & 18	34.7V±1%; 30.1 ohms, ±1%, 10W	17.35V ± 2%
11 & 18	67.1V±1%; 15.0 ohms, ±1%, 75W	33.5V ± 2%
2 & 4	259.8V±1%; 100 ohms, ±1%, 75W	173.2V± 2%
3 & 4	259.8V±1%; 100 ohms, ±1%, 75W	173.2V ± 2%

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.

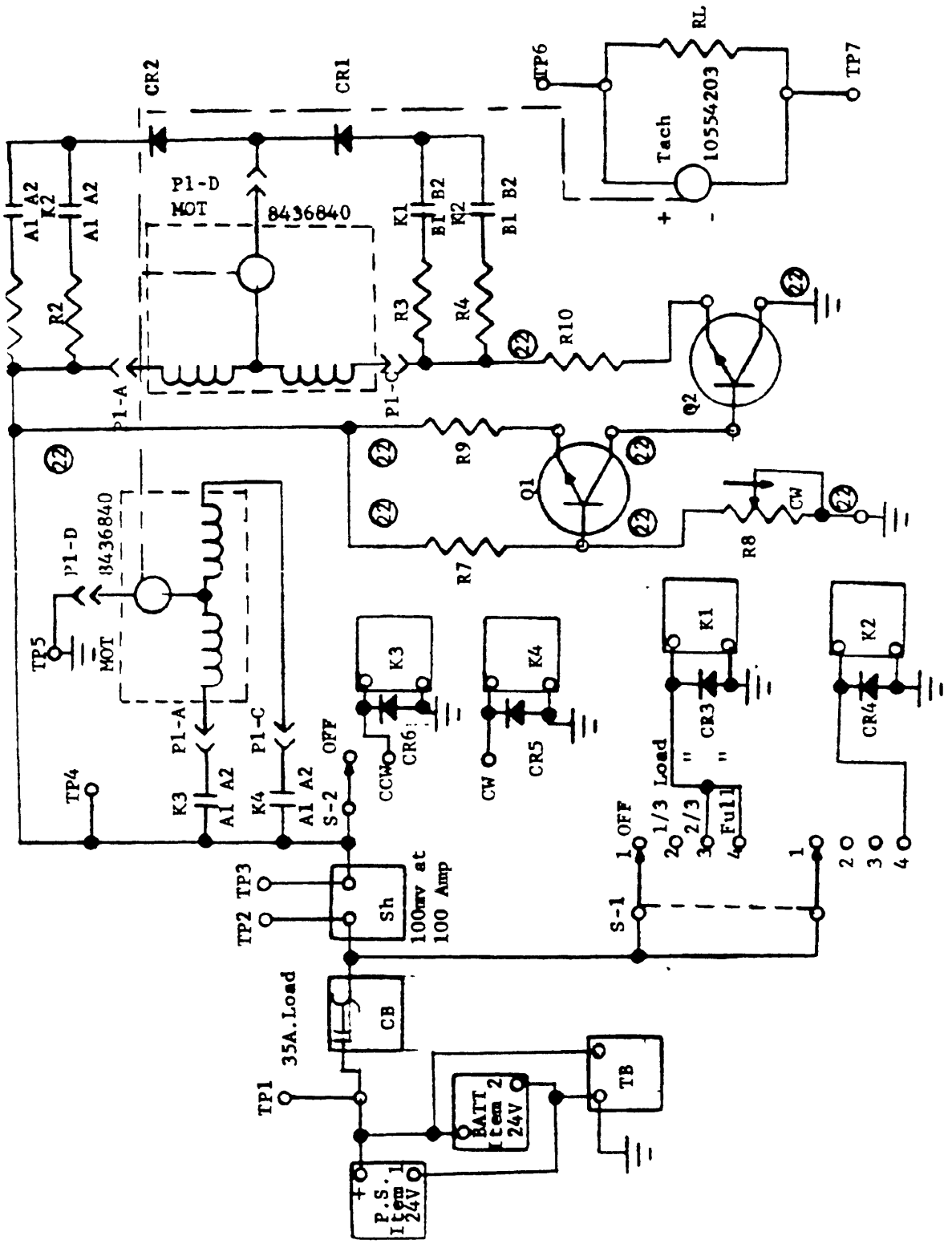


FIGURE 1. MOTOR ASSEMBLY TEST FIXTURE

Source: https://assist.dla.mil -- Downloaded: 202640-22T19:33Z
Check the source to verify that this is the current version before use.

3.4.2.3 Intermittent stitching charge maintenance circuit.

3.4.2.3.1 Conditions of operations.- Unless otherwise stated, the requirements of 3.4.2.3 shall be met with the test configuration shown in Figure 2.

3.4.2.3.2 Charge maintenance.- After the 29.0 ± .5 Vdc has been removed from between J1-29 and J1-32, the relay shall maintain contact l t Al for l t least 0.2 seconds. See Figure 2.

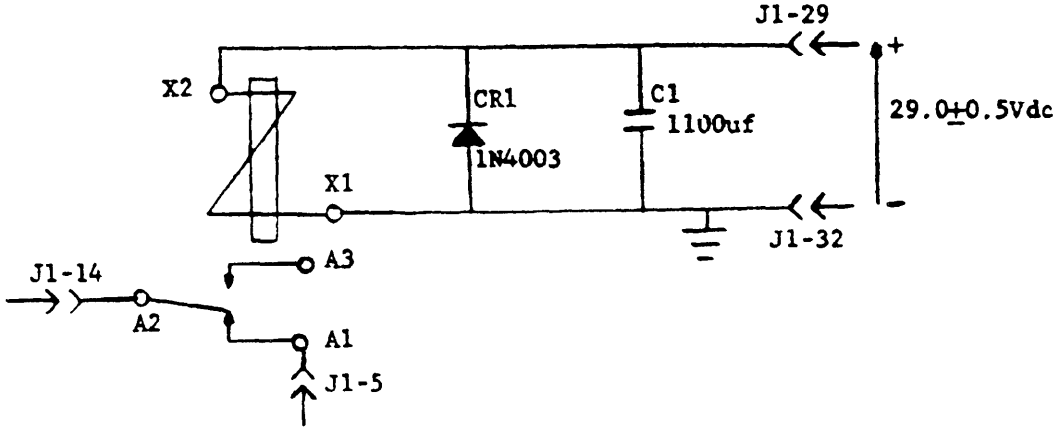


FIGURE 2

3,4.3 Wire (hookup).- Standard conductors of suitable degrees of flexibility shall be used whenever practicable within the requirements of this specification. hookup tire l hall be of such cross section and temper as to provide ample and safe current-carrying capacity and mechanical strength. In general, wire shall not be smaller than AWC No. 22. Wire directly associated with the primary power circuit shall be of a size consistent with the current requirements of the circuit. When it can be demonstrated that the use of smaller hookup wire will result in no loss of performance and at the same time will produce benefits, such as reduction in weight or improved l ccessibility, smaller wire may be used, This provision is made specifically to permit the we of smaller wire in cables having l large number of wires and having adequate support against l dverse l ffacts of vibration. The size of wire leads supplied with parts shall be controlled by the l pecifica-tions for these parts.

3.4.3.1 Insulation on conductors.- Insulation of the sleeve type may be used, provided its length is sufficient to prevent slippage l long the conductor to such an extent as to expose l l ection of the conductor that should be Insulated, Insulation shall provide l dequate dielectric l trength and adequate leakage resistance when the equipment is operated under the designated l e-ce conditions. The mechanical and electrical properties of the insulating materials and impregnating compounds shall not be permanently Impaired as a result of being exposed to extremes of temperature incurred in the equipment. Rubber without braid may be used only when the insulation is not depended upon to space or support the conductor, or when l number of conductors

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:33Z
Check the source to verify that this is the current version before use.

are assembled in the form of cordage l nd the assembly provided with a braid covering which has been suitably varnished or lacquered, All conductor insulation shall be of the noncombustible or slow-burning type; that is, with the insulated conductor held in a horizontal position in still air, self-sustained combustion of the insulation or lacquer shall not progress at a rate in excess of 1 inch in 1 minute.

4. QUALITY ASSUREANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the suppiler is responsible for the performance of all inspection requirements, as specified herein, Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article (initial production) approval.- The requirements for first article approval l nd the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall be 3 chassis assemblies manufactured in the same manner using the same materials, equipment, processes, and procedures as used in regular production, All parts and materials ‘including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 government testing .- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government shall inspect the sample to insure that it conforms to all the requirements of the contract and submit l record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and l record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor’s inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation l nd presentation of lots shall be in accordance with “Submission of Products” as specified in MXL-STD-105.

4.3.2 Examinations and tests.- Except where otherwise specified herein, components and subassemblies shall be inspected in accordance with the “Inspection provisions for components and subaso-lice” l nd the Sampling Plan Tables contained in MIL-F-13926. Examinaions l nd

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.

teats related to Section 3 herein shall be perfomed in accordance with Tables III and IV and test methods and procedures in 4.5. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein.

4.3.2.1 Components and subassemblies.- Printed circuit subassemblies shall be inspected in accordance with the inspection provisions of MIL-C-60898. All other components and subassemblies shall be inspected to the applicable requirements of MIL-F-13926.

4.3.2.2 Control tests.- One chassis assembly shall be selected at random as a control sample from each 100 produced. The control samples shall meet the requirements and tests of Table III after they have met the requirements and tests of Table IV.

TABLE III

TEST	REQUIREMENTS	TEST PROCEDURES
Fabrication	3.1	Applicable drawings-visual
General specifications	3.2	Visual-tactile
Vibration	3.3*1	4.5.1.1, 4.5.1.2
Temperature	3.3.2, 3.4.3	4.5.1.1, 4.5.1.4
Humidity	3.3.3	4.5.1.1, 4.5.1.3
Shock	3.3*4	4.5.1.1, 4.5.1.5
Altitude	3.3.5	4.5.1.1, 4.5.1.6
Sand and dust	3.3.6	4.5.1.1, 4.5.1.7
Salt Spray	3.3.7	4.5.1.1, 4.5.1.8
Fungus	3.3,8	4.5.1.1, 4.5.1.9

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with “Acceptance and Rejection” as specified in MIL-STD-105.

4.3.4 Functional tests.- The requirements and teats in Table IV shall be inspected on a 100% basis. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+50° to +90°F
b,	Relative humidity	90% RH max.
c,	Pressure	28 to 32 inches of Hg

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:33Z
Check the source to verify that this is the current version before use.

'TABLE IV

(100 Percent Inspection)

Characteristic	Requirement	Procedure
101, Continuity Test	3.4.1	4.5.2.1
102. Motor Assembly Conditions of Operation	3.4.2.1.1	4.5.2.2.1.1
103. CCW output	3.4.2.1.2	4.5.2.2.1.2
104, CW output	3.4.2.1.3	4.5.2.2.1.3
105, Resistor Loads Conditions of Operations	3.4.2.2.1	4.5.2.2.2.1
106. Resistive Loads Output	3.4.2.2.2	4.5.2.2.2.2
107, Intermittent Switching Charge Maintenance Circuit Conditions of Operations	3.4.2.3.1	4.5.2.2.3.1
108. Charge Maintenance	3.4.2.3.2	4.5.2.2.3.2
109. Relay and Diode Resistance	3.4.2.3.3	4.5.2.2.3.3

4.4 Inspection equipment,- Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component Inspection" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of tent equipment is normally . considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus , concept, construction, materials, dimensions l nd tolerances used in the design of test equipment shall be so selected l nd controlled as to insure that the test equipment will reliably Indicate acceptability of a product which does not exceed 90% of the prescribed tolerance Limit, and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.

4.4,1.1 Test equipment .- In conjunction with 4.4.1, the following test equipment shall be utilized in the performance of the applicable tests as specified in 4,5:

ITEM		DESCRIPTION
1.	Power supply	24±.5Vdc, 20 amps, load regulation of ±0.5%, less than .1% ripple.
2.	Storage Battery	D8437166
3.	Battery Charger	B10556495
4.	Motor Assembly Test Harness	Table V and Figure 1
5.	Digital Multimeter	D.C. Voltmeter: O-300V, 0.17. accuracy, input impedance shall be 1 megohms or greater. Ohmmeter: O to lK ohms, .2% l ccuracy.
6.	Power Supply	0-300 Vdc adjustable, O-3 amps, 100 mv resolution, load regulation, of ±0.5%, less than .1% ripple.
7.	Analog voltmeter	0-50 Vdc 2% accuracy
8.	Ammeter	0-200 ma dc, ±1% accuracy
9.	Teat Resistor Harness	Table VI and Figures 3 thru 13
10,	Stitch (S-3)	8437461-2

Source: https://assist.dla.mil -- Downloaded: 202640-22T19:33Z
Check the source to verify that this is the current version before use.

TABLE V

MOTOR ASSEMBLY TEST HARNESS, ITEM 4

DESIGNATION	PART	DESCRIPTION
MOT	2 Motors, Drive	D8436840
Tach	Tachometer Assembly	10554203
TB	Terminal Board	
CB	Circuit Breaker	MS25244-35
Sh	Shunt	10554229
S-1	Switch	10554155-6
S-2	Switch	MS22885/9-004
K1, K2, K3, K4	4 Relays	MS24376Dl
R1, R2, R3, R4	4 Resistors	MIL-R-18546D, RE80GR499
R9	Resistor	MIL-R-l8546D, RE65G84R5
R10	Resistor	MIL-R-18546D, 10554118-1
RL	Resistor	100k ohms, ± 5%, .1 watt
CR1, CR2	2 Diodes	1N1184
CR5, CR6	2 Diodes	1N4003, 10545512
Q2	Transistor	**10554199**
TP1	Component Assembly	10554200
TP1 thru TP7	7 Tip Jacks	h.h Smith or equiv.
P1 & P2	2 connectors	MS-3106F20-4S
--	Wire (hookup, Figure 1, all branches where encircled 22 is found)	AWG-Z2 Gauge
--	Wire (hookup, Figure 1, all branches where no encircled 22 is found)	AWG-12 Gauge

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:33Z
Check the source to verify that this is the current version before use.

TABLE VI

TEST RESISTOR HARNESS, ITEM 9

DESIGNATION	PART	DESCRIPTION
RT 13	Resistor	MIL-R-18546D, RE75N14R0
RT 1	Resistor	MIL-R-18546D, RE75N14R0
RT 2	Resistor	MIL-R-18546D, RE70N6490
RT 5	Resistor	MIL-R-18546D, RE75N6980
RT 9	Resistor	MIL-R-18546D, RE75NllR8
RT 10	Resistor	MIL-R-18546D, RE75N11R8
RT 4	Resistor	MIL-R-18546D, RE75N4020
RT 14	Resistor	MIL-R-18546D, RE75N4020
RT 3	Resistor	MIL-R-18546D, RE75N4020
RT 15	Resistor	MIL-R-18546d, RE75N4020
RT 11	Resistor	MIL-R-18546D, RE65N30RI
RT 12	Resistor	MIL-R-18546D, RE77N15R0
RT 6	Resistor	MIL-R-18546D, RE77NI000
RT 7	Resistor	MIL-R-18546D, RE77Nl000

4.5 Test methods and procedures.

4.5.1 Environmental.

4.5.1.1 Test conditions.- The chassis assembly shall be mounted in an upright position in a manner similar to that in which it will be in service and shall be subjected to the environmental tests of the following subparagraphs. The l tmospheric pressure shall be room ambient. Except for the high and low temperature tests or the humidity tests, as l pplicable, the temperature and humidity shall be room ambient.

4.5.1.2 Vibration.- The vibration test shall be made on a device with means provided for measuring l nd controlling the direction of vibration and adjusting and amplitudes and frequencies to keep them within the specified limits of 3.3.1. The measuring instruments of the testing device shall be capable of measuring the vibration amplitude within plus or minus 5 percent, and capable of measuring the frequency within plus or minus 2 percent. The test shall be performed in accordance with the requirements specified in 3.3.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.

Subsequent to the vibration test, the test shall show no damage or defect resulting from the test when examined by visual and tactile means. The chassis assembly shall then be tested to determine compliance with requirements specified under 3.3.1.

4.5.1.3 Humidity.- The humidity test shall be performed in accordance with Procedure 1 of Method 507 in MIL-STD-81O. Subsequent to the humidity test, the shassis l ssembly shall be tested to determine compliance with 3.3.3.

4.5.1.4 Temperature. - Test in accordance with Procedure 1 of Method 501 in MIL-STD-81O for high temperature operating and nonoperating under conditions specified in 3.3.2. Test in accordance with Procedure 1 of Method 502 for low temperature operating l nd nonoperating under conditions specified in 3.3.2, and verify requirements of 3.3.2.1 and 3.3.2.2.

4.5.1.5 Shock.- The l hock test shall be performed in l ccord.anca with MIL-T-5422.

4.5.1.6 Altitude.- The altitude test shall be performed in accordance with the following steps:

Step 1. Place the chassis assembly in the test chamber in l manner that will simulate service usage, making connections and attaching instrumentation as necessary. Plugs, covers, and inspection plates not used in operation, but used in servicing shall remain In place. When mechanical or electrical connections are not used, the connections normally protected in service shall be adequately covered. For tests where temperature values are controlled, the test chamber shall be at standard ambient conditions when the chassis asseably is installed. The chassis assembly shall then be operated to detemnine “that no malfunction or damage was caused due to faulty installation or handling. The requirement for operation following installation of the chassis assembly in the test facility is l pplicable only when opertion is required during or immediately following exposure to the specified test. Standard ambient temperature shall be maintained during the entire test.

“- -step 2. Reduce the chamber internal pressure to 87.5mm of Hg (3.44 In. of hg or 50,000 ft. above sea level). The rate of pressure change may be the maximum attainable by the chamber. Maintain the chamber pressure for a period of not less than one hour.

Step 3. Increase the chamber pressure to 523mm of Hg (20.6 in. of Hg or 10,000 ft. above sea level) l nd then operate the shassis assembly and compare the results with the data obtained in accordance with pretest performance record nude of all data necessary to determine compliance with required performance.

Step 4. With the chassis assembly not operating, return the chamber to standard ambient conditions,

Step 5. Operate the chassis assembly l nd compare the data obtained with the data obtained in accordance with pretest performance record.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.

Step 6. The chassis assembly shall be visually inspected and a record made of any damage or deterioration resulting from the test. If a test chamber is used for the test, perform a visual inspection of the test within the chamber at test conditions, when possible. Upon completion of the test, visually inspect the chassis assembly again after the chassis assembly has been returned to standard ambient conditions. Deterioration, corrosion or change in tolerance limits of any internal or l xternal parts which in l ny manner prevent the chassis assembly from meeting operational service or maintenance requirements shall provide reason to consider the chassis assembly as hating failed to withstand the conditions of the test.

4.5.1,7 Sand and dust.- The sand and dust test shall be performed in accordance with MIL-T-5422.

4.5.1.8 Salt spray.- The salt spray test shall be performed in accordance with MIL-T-5422.

4.5.1.9 Fungus.- The fungus resistance test shall be performed in accordance with MIL-T-5422.

4.5.2 Electrical tests,

4.5.2.1 Continuity test.- A continuity test shall be performed to verify conformance to 3.4.1.

4.5.2.2 Functional tests.

4.5.2.2.1 Motor assembly

4,5.2.2.1.1 Conditions of operation.- Connect Power Supply (item 1) and Storage Battery (item 2) to test fixture as shown in Figure 1. Check voltage at TP1 with D.C. Voltmeter (item 5) to insure It is 24±.5Vdc.

4.5,2.2.1.2 Counterclockwise output.- Place switch S-2 in CCW position. Use Digital Voltmeter (item 5) to measure and monitor D.C. voltages at Servo Load (Tp-2, Tp-3), Tach (Tp-2, Tp-3), and Tp-4, Tp-5 test points respectively for 1/3, 2/3, and full load conditions l s specified in Table I to insure conformance to 3,4.2.1.2.

4.5.2.2.1.3 Clockwise output.- clockwise output testing shall be performed as specified in 4.5.2.2.1.2 except that Switch S-2 shall be set to CW position. Measured and monitored voltages shall conform to requirements of 3.4.2.1.3.

4.5.2.2.2 Resistor loads.

4.5.2.2.2.1 Conditions of operation.- Connect Power Supply (Item 6), Digital D.C. Voltmeter (Item 5) and Test Resistor Harness (Item 9) to Resistor load terminals as shown in Table VII and Figures 3 through 13 respectively.

4.5.2.2.2.2 Resistor load outputs.- Adjust Power Supply (Item 6) and read D.C. Output Voltage on D.C. Voltmeter (Item 5) for each J1 terminal pair specified in Table VII to insure conformance to 3.4.2.2.2,

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.

TABLE VII

J1 Terminal	Figure	Power Supply Setting in Volts D.C.	Output Voltage In Volts D.C.
26 & 32	3	41.0 ± 1%	20.5 ± 2%
21 & 32	4	227.9 ± 1%	113.9 ± 2%
6 & 27	5	289.4 ± 1%	144.7 ± 2%
16 & 32	6	37.6 ± 1%	18.8 ± 2%
19 & 32	7	37.6 ± 1%	18.8 ± 2%
1 & 32	8	219.6 ± 1%	109.8 ± 2%
22 & 32	9	219.6 ± 1%	109.8 ± 2%
12 & 18	10	34.7 ± 1%	17.35 ± 2%
11 & 18	11	67.1 ± 1%	33.5 ± 2%
2 & 4	12	259.8 ± 1%	173.2 ± 2%
3 & 4	13	259.8 ± 1%	173.2 ± 2%

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:33Z
Check the source to verify that this is the current version before use.

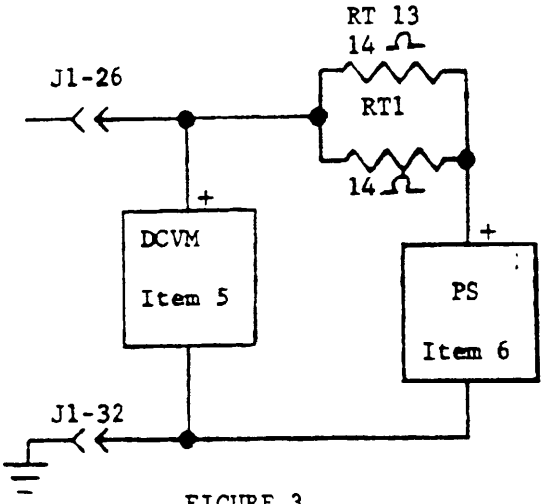


FIGURE 3

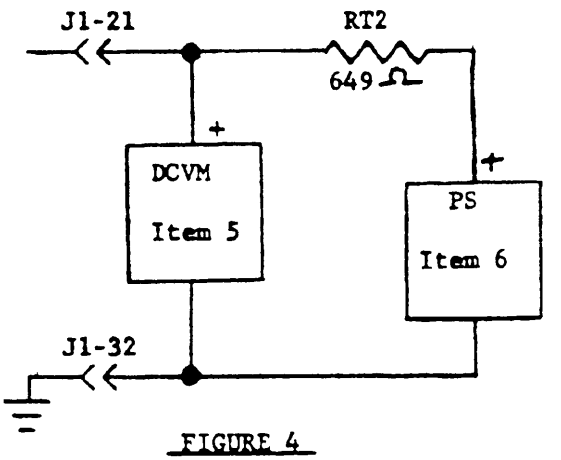


FIGURE 4

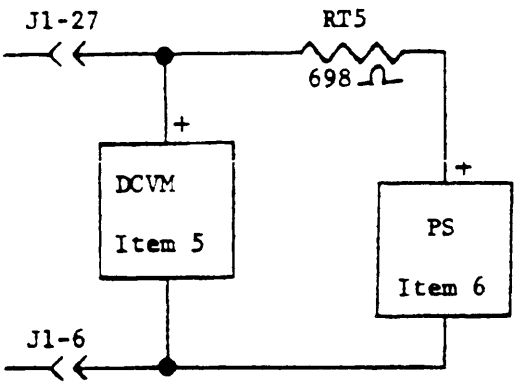


FIGURE 5

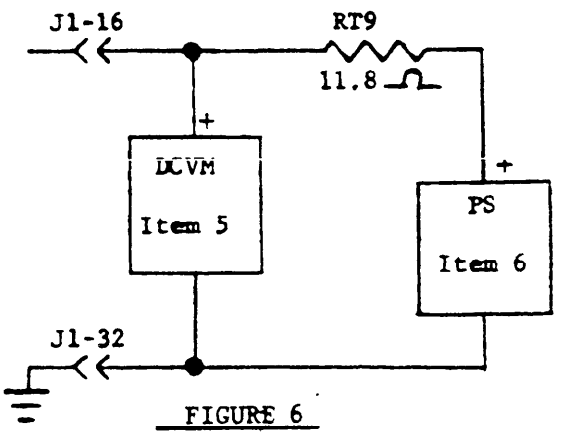


FIGURE 6

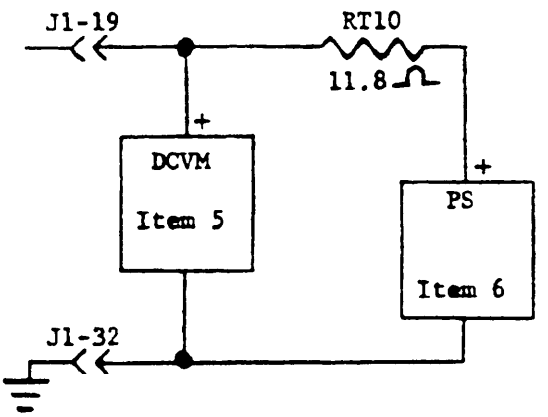


FIGURE 7

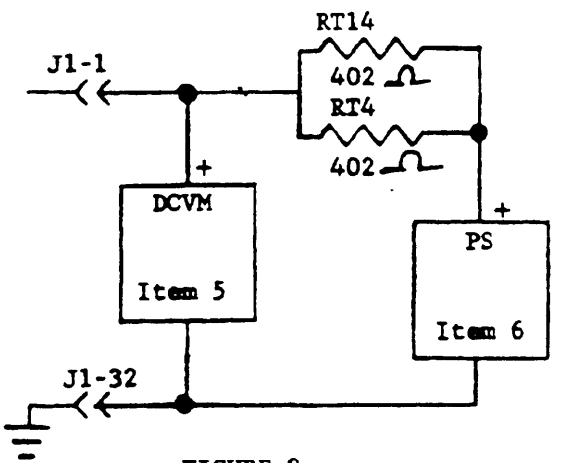


FIGURE 8

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:33Z
Check the source to verify that this is the current version before use.

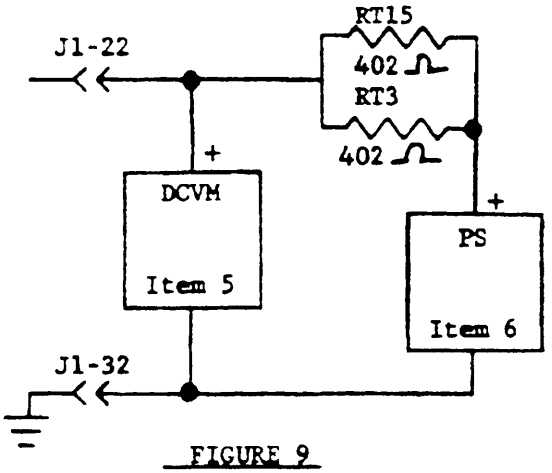


FIGURE 9

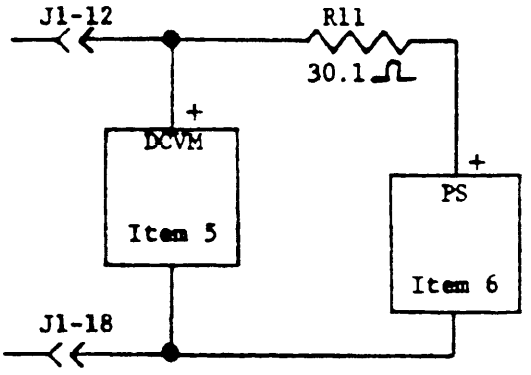


FIGURE 10

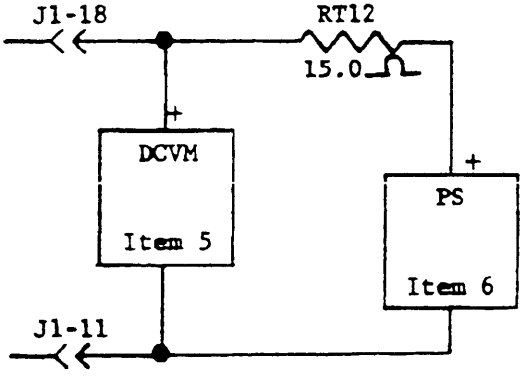


FIGURE 11

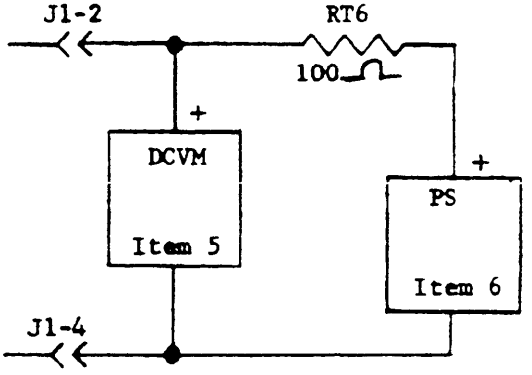


FIGURE 12

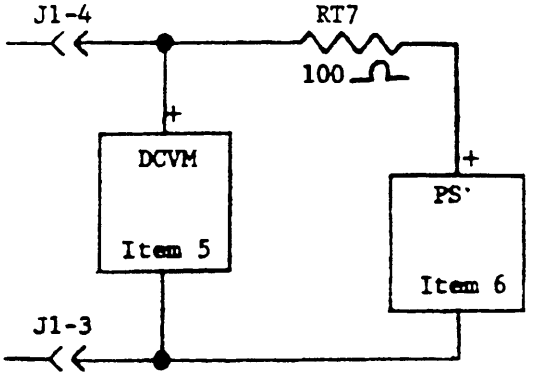


FIGURE 13

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.

4.5.2.2.3 Intermittent switching charge maintenance circuit.

4.5.2.2.3.1 Conditions of Operations. - Connect Power Supply (Item 6) and Ammeter (Item 8) through Stitch S-3 (Item 10) to J1-29 terminal and Analog Voltmeter (Item 7) between J1-14 and J1-5 terminals l t shown in Figure 14.

4.5.2.2.3.2 Charge maintenance Adjust Power Supply to 29.0 ± 0.5 Vdc and turn Switch to ON position, the Ammeter shall read 150 ma ± 10%, and the Analog Voltmeter shall indicate continuity. Turn Switch to OFF position l nd the Analog Voltmeter shall indicate the relay is making contact at Al l nd A2 for l t least 0.2 seconds after the Switch had been turned to the OFF position.

4.5.2,2.3.3 Relay l nd Diode Resistances.- With the Switch tuned to OFF position, connect the + lead of the digital D.C. Voltmeter (Item 5) to the cathode side of the diode (JI-29) and the common lead to J1-32 terminal, the Digital D.C. Voltmeter (Item 5) shall read the relay resistance of 193 ohms ± 2%, Reverse the leads of the Digital D.C. Voltmeter (Item 5), and read the diode resistance of 5 ± 1 ohms.

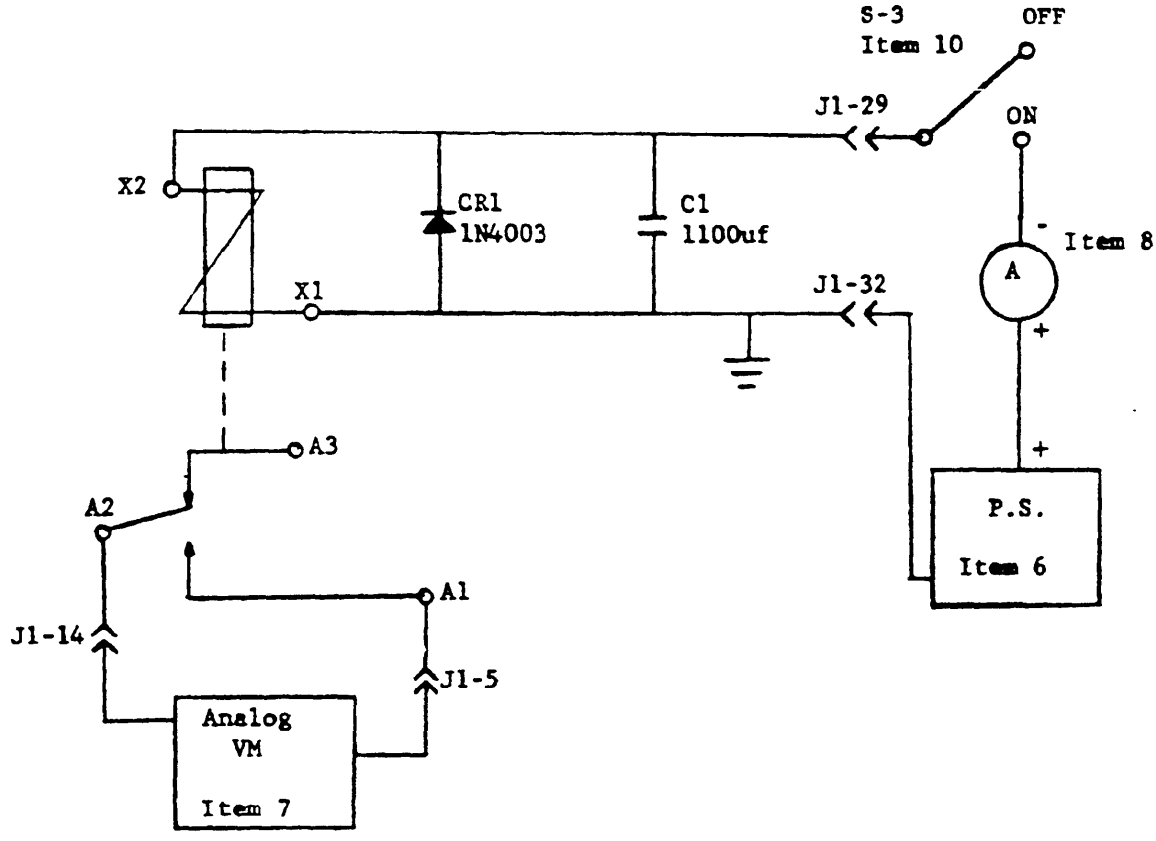


FIGURE 14

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.

4.5.3 Wire insulation.- Wire insulation requirements shall be verified by certification.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking.- Preservation packaging, packing, and marking shall be in accordance with Packaging Data Sheet 11748238.

6. NOTES

6.1 Intended use.- This chassis assembly is designed for rackmounting into the AN/TSM-100 Electronics System Test Set, which is used for troubleshooting and performance checking of l pacific weapon system components.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number l nd data of this Specification.

b. Applicable stock number.

co A block of serial numbers

d. Instructions for first article inspection.

e. Level of protection required.

Custodian

Army-MU

Preparing Activity:

Amy-Mu

Project No. 4931-A140

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:33Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
Frankford Arsenal
ATTN: SARFA-MDM
Philadelphia, PA 19137

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22-R255

INSTRUCTIONS The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE MIL-T-48587(MU) Test Set, Inverter, Converter and Motor Chassis Assembly: 11748238

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 202640-22T19:33Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.